Scalability & Future Plan

Date	5 October 2026
Team ID	SWTID-2026-5240
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	5 Marks

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited concurrent user handling (Single server)	System slows down / crashes when many users access at same time	High
2	Local file-based / limited database storage	Not suitable for large datasets, risk of data	High
3	No advanced caching and load balancing	High response time and poor user experience	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles ~100 users on single instance	Deploy on Cloud (AWS/Azure) with Auto Scaling, Load Balancer, Docker + Kubernetes
2	Data Storage	Local / SQLite storage	Migrate to MongoDB / Firebase / AWS RDS, use S3 for files
3	Performance	No caching, direct DB queries	Implement Redis caching, CDN, API optimization & indexing
4	Security	Basic authentication	JWT/OAuth2, HTTPS, encryption, role-based access, vulnerability scanning

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact

Phase 2	Mobile App Integration + Real-time Notifications + Multi-language	3-6 Months	Increased user reach and engagement
Phase 3	AI/ML Model Enhancement + Predictive Analytics + Chatbot	6-12 Months	Smarter decisions & automation
Phase 4	IoT Integration + Global Deployment + Monetization Model	12+ Months	Large-scale adoption & commercial readiness